

MSj01 Project

Frezza metabolomics

Batch 1

Previously extracted samples.

MSJ01_HG_003	HG2_30min_Risk	heart	Naked mole-rat (HG)
MSJ01_HG_006	HG3_30min_Adjacent	heart	Naked mole-rat (HG)
MSJ01_HG_007	HG4_30min_Risk	heart	Naked mole-rat (HG)
MSJ01_HG_009	HG1_5min_Risk	heart	Naked mole-rat (HG)
MSJ01_HG_010	HG1_5min_Adjacent	heart	Naked mole-rat (HG)
MSJ01_HG_013	HG3_5min_Risk	heart	Naked mole-rat (HG)
MSJ01_HG_014	HG3_5min_Adjacent	heart	Naked mole-rat (HG)
MSJ01_MM_022	MM_F2_Sham	heart	Mouse Female (MM_F)
MSJ01_MM_029	MM_F1_5min_Risk	heart	Mouse Female (MM_F)
MSJ01_MM_032	MM_F2_5min_Adjacent	heart	Mouse Female (MM_F)
MSJ01_MM_034	MM_F3_5min_Adjacent	heart	Mouse Female (MM_F)
MSJ01_MM_035	MM_F4_5min_Risk	heart	Mouse Female (MM_F)
MSJ01_MM_037	MM_M1_5min_Risk	heart	Mouse Male (MM_M)
MSJ01_MM_038	MM_M1_5min_Adjacent	heart	Mouse Male (MM_M)
MSJ01_MM_039	MM_M2_5min_Risk	heart	Mouse Male (MM_M)
MSJ01_MM_040	MM_M2_5min_Adjacent	heart	Mouse Male (MM_M)
MSJ01_MM_041	MM_M3_5min_Risk	heart	Mouse Male (MM_M)
MSJ01_MM_043	MM_M4_5min_Risk	heart	Mouse Male (MM_M)
MSJ01_MM_044	MM_M4_5min_Adjacent	heart	Mouse Male (MM_M)
MSJ01_MM_045	MM_F1_30min_Risk	heart	Mouse Female (MM_F)
MSJ01_MM_047	MM_F2_30min_Risk	heart	Mouse Female (MM_F)
MSJ01_MM_049	MM_F3_30min_Risk	heart	Mouse Female (MM_F)
MSJ01_MM_050	MM_F3_30min_Adjacent	heart	Mouse Female (MM_F)
MSJ01_MM_051	MM_F4_30min_Risk	heart	Mouse Female (MM_F)
MSJ01_MM_053	MM_M1_30min_Risk	heart	Mouse Male (MM_M)
MSJ01_MM_055	MM_M2_30min_Risk	heart	Mouse Male (MM_M)
MSJ01_MM_057	MM_M3_30min_Risk	heart	Mouse Male (MM_M)
MSJ01_MM_059	MM_M4_30min_Risk	heart	Mouse Male (MM_M)

Batch2

Frozen tissue received.	MSJ01_HG_001_b	HG1_30min_Risk	heart	Naked mole-rat (HG)
	MSJ01_HG_002_b	HG1_30min_Adjacent	heart	Naked mole-rat (HG)
	MSJ01_HG_003_b	HG2_30min_Risk	heart	Naked mole-rat (HG)
	MSJ01_HG_004_b	HG2_30min_Adjacent	heart	Naked mole-rat (HG)
	MSJ01_HG_005_b	HG3_30min_Risk	heart	Naked mole-rat (HG)
	MSJ01_HG_006_b	HG3_30min_Adjacent	heart	Naked mole-rat (HG)
	MSJ01_HG_007_b	HG4_30min_Risk	heart	Naked mole-rat (HG)
	MSJ01_HG_008_b	HG4_30min_Adjacent	heart	Naked mole-rat (HG)
	MSJ01_HG_010_b	HG1_5min_Adjacent	heart	Naked mole-rat (HG)
	MSJ01_HG_011_b	HG2_5min_Risk	heart	Naked mole-rat (HG)
	MSJ01_HG_012_b	HG2_5min_Adjacent	heart	Naked mole-rat (HG)
	MSJ01_HG_013_b	HG3_5min_Risk	heart	Naked mole-rat (HG)
	MSJ01_HG_014_b	HG3_5min_Adjacent	heart	Naked mole-rat (HG)
	MSJ01_HG_016_b	HG4_5min_Adjacent	heart	Naked mole-rat (HG)
	MSJ01_HG_017_b	HG1_Sham	heart	Naked mole-rat (HG)
	MSJ01_HG_018_b	HG2_Sham	heart	Naked mole-rat (HG)
	MSJ01_HG_019_b	HG3_Sham	heart	Naked mole-rat (HG)
	MSJ01_HG_020_b	HG4_Sham	heart	Naked mole-rat (HG)
	MSJ01_MM_021_b	MM_F1_Sham	heart	Mouse Female (MM_F)
	MSJ01_MM_023_b	MM_F3_Sham	heart	Mouse Female (MM_F)
	MSJ01_MM_024_b	MM_F4_Sham	heart	Mouse Female (MM_F)
	MSJ01_MM_025_b	MM_M1_Sham	heart	Mouse Male (MM_M)
	MSJ01_MM_026_b	MM_M2_Sham	heart	Mouse Male (MM_M)
	MSJ01_MM_027_b	MM_M3_Sham	heart	Mouse Male (MM_M)
	MSJ01_MM_028_b	MM_M4_Sham	heart	Mouse Male (MM_M)
	MSJ01_MM_031_b	MM_F2_5min_Risk	heart	Mouse Female (MM_F)
	MSJ01_MM_033_b	MM_F3_5min_Risk	heart	Mouse Female (MM_F)
	MSJ01_MM_037_b	MM_M1_5min_Risk	heart	Mouse Male (MM_M)
	MSJ01_MM_038_b	MM_M1_5min_Adjacent	heart	Mouse Male (MM_M)
	MSJ01_MM_043_b	MM_M4_5min_Risk	heart	Mouse Male (MM_M)
	MSJ01_MM_044_b	MM_M4_5min_Adjacent	heart	Mouse Male (MM_M)
	MSJ01_MM_054_b	MM_M1_30min_Adjacent	heart	Mouse Male (MM_M)
	MSJ01_MM_056_b	MM_M2_30min_Adjacent	heart	Mouse Male (MM_M)
	MSJ01_MM_058_b	MM_M3_30min_Adjacent	heart	Mouse Male (MM_M)
	MSJ01_MM_059_b	MM_M4_30min_Risk	heart	Mouse Male (MM_M)

Issues

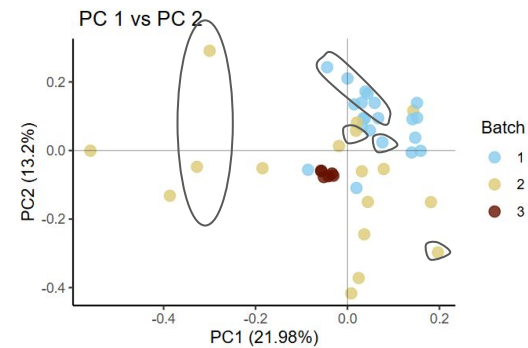
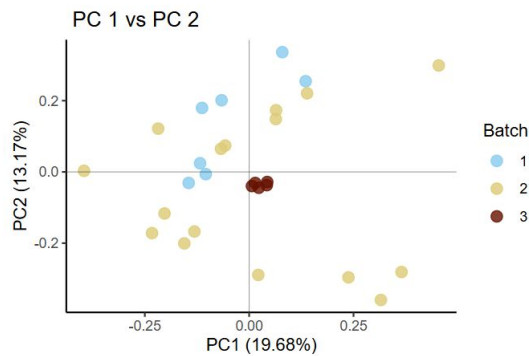
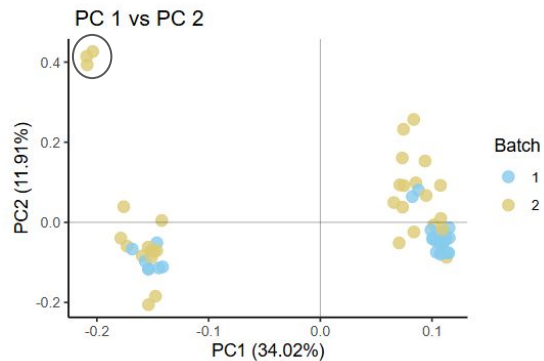
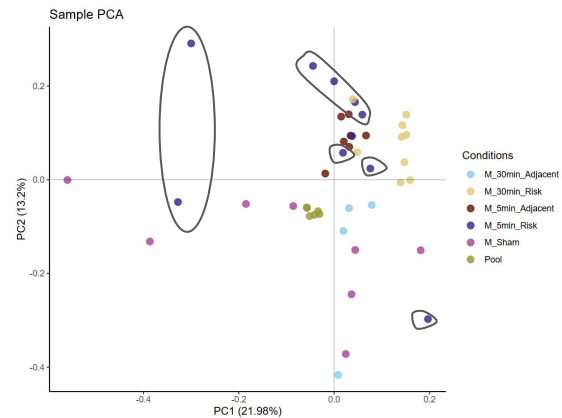
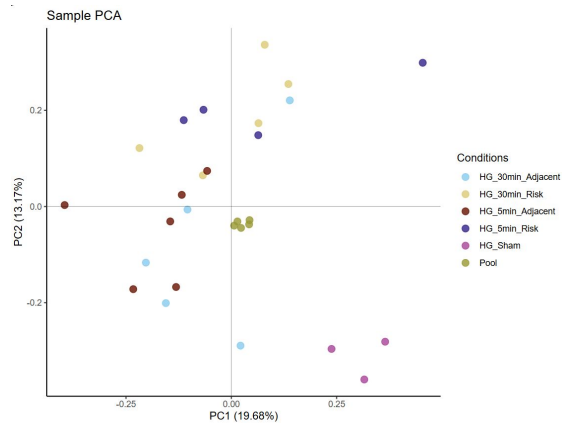
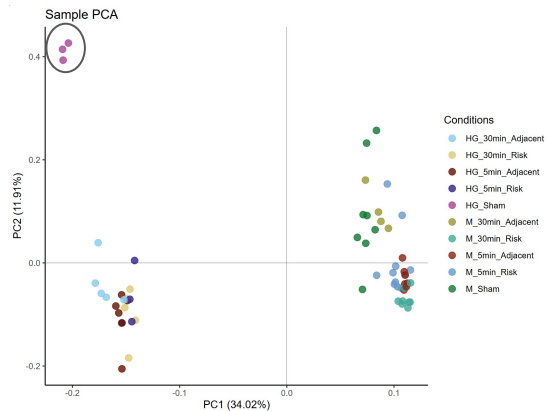
- Unable to assess/Correct Batch effects
 - Issues with Sample grouping/ NO clear trajectories in PCA
 - Not all sample groups exist in both batches
 - NMR CTRL group is only in Batch 2

PCA/Batch

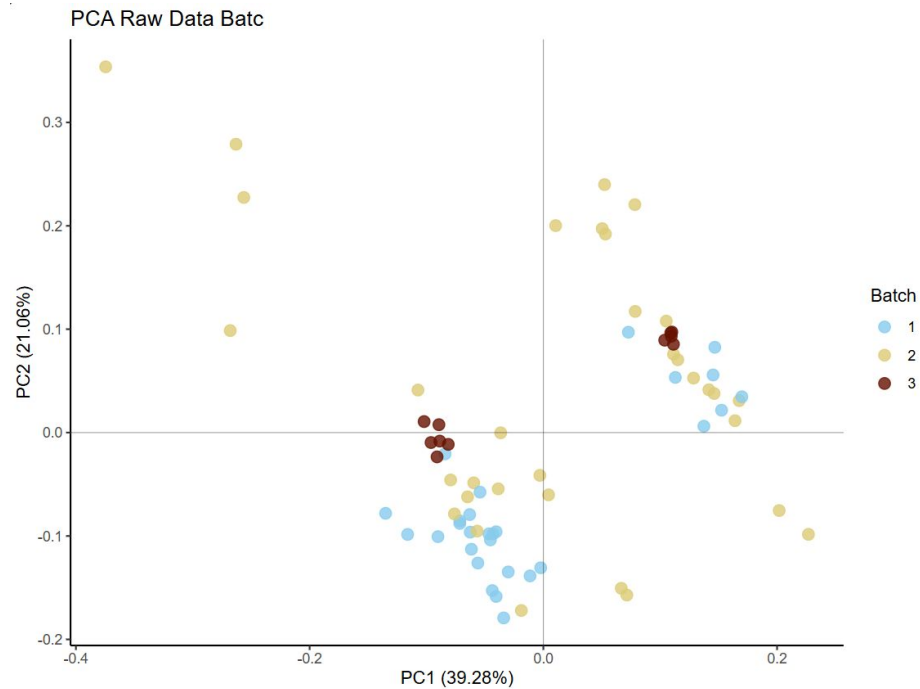
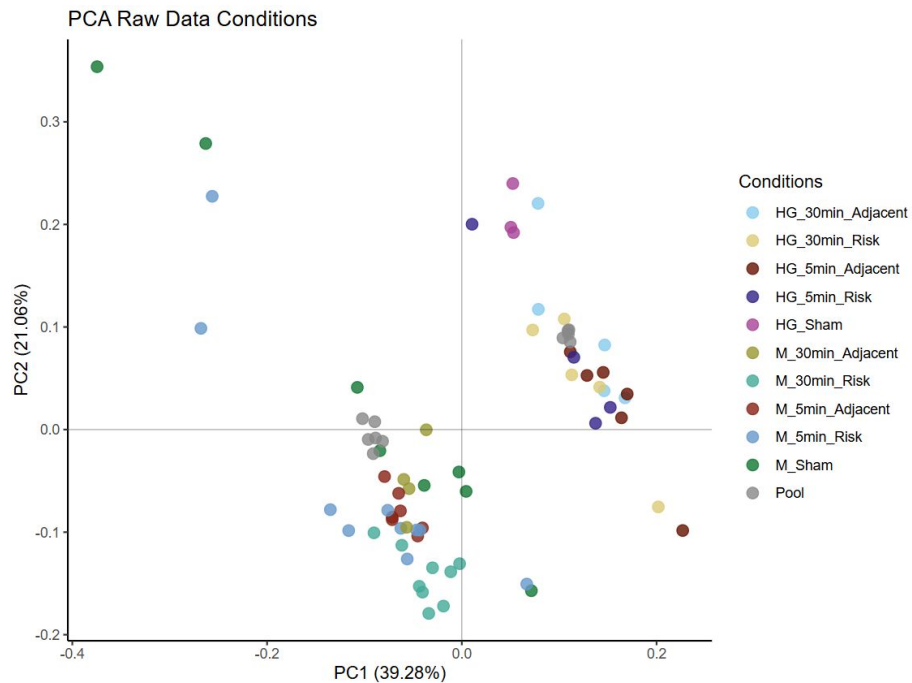
HG only

MM only

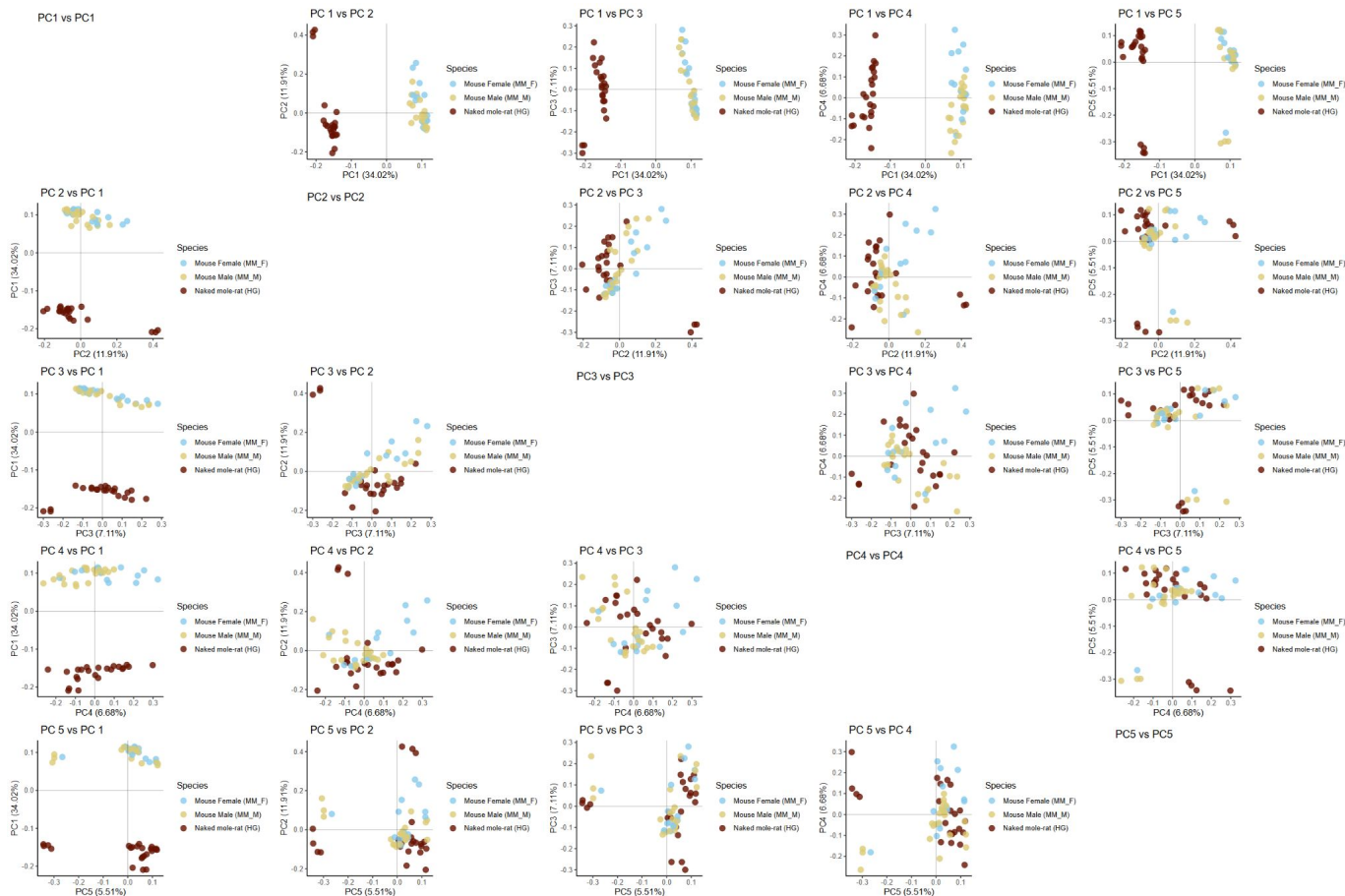
Together



Batch raw



No clear sex dependency in mouse



Check some known metabolites for Experimental validity

Seem ok!

